

Curriculum Vitae

Chao Tan

Ph.D. Student in Electrical and Computer Engineering

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Profile

Ph.D. student at The University of Hong Kong advised by [Prof. Edmund Y. Lam](#). My research centers on efficient and rapid perception across spatial scales, from macroscopic scenes to microscopic structures, and across representations from 2D to 3D. I develop physics-informed imaging systems and algorithms spanning event-based vision, computational photography, and computational microscopy, with the long-term goal of capturing and recreating meaningful memories as rich, multidimensional experiences beyond conventional photographs and videos.

Education

The University of Hong Kong

Sep 2026 – Present

Doctor of Philosophy in Electrical and Computer Engineering

- **Advisor:** [Prof. Edmund Y. Lam](#)
- **Research Fields:** Computational imaging, machine perception, event-based vision

Korea Advanced Institute of Science and Technology

Sep 2024 – Aug 2026

Master of Science in Bio and Brain Engineering

- **Advisor:** [Prof. Mooseok Jang](#)
- **Research Fields:** Fourier ptychographic tomography, computational microscopy
- **Selected Courses:** Computational Biomedical Optics, Machine Learning for 3D Data, Interactive Computer Graphics

Sichuan University

Sep 2020 – Jun 2024

Bachelor of Engineering in Optoelectronic Information Science and Engineering

- Major in Architecture (Sep 2020 – Jun 2021)

- **GPA:** 3.61 / 4.0 (Compulsory Courses: 3.68 / 4.0)

Selected Awards

KAIST International Student Scholarship

2024 – 2026

- Full tuition coverage + monthly living stipend

First Prize for Outstanding Undergraduate Thesis

2024

Sichuan University, College of Electronics and Information Engineering

- **Thesis:** Research on Computer-generated Holography and Fourier Ptychographic Microscopy Based on Diffraction Calculation
- Ranked among the top 3 theses in the college

Science Innovation Program - Key Project

2023

Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences

- **Project:** Depth-of-Field Extension via Fourier Ptychography
- Awarded ¥20,000 as top-tier competitive grant

Publications

([†] indicates equal contribution;  indicates corresponding author.)

- [1] **Chao Tan**, F. Lu, S. Park, H. Na, C. Lee, C.-S. Kim, J. Kim, H. Lee, and M. Jang, “Feature-domain Fourier ptychographic tomography with dark-field illumination”, *arXiv:2608.21890* (2026).
- [2] H. Na, S. Lee, **Chao Tan**, D. Lee, and M. Jang, “Beyond the weak-scattering limit: accurate Kramers–Kronig phase imaging of high-contrast reflective surfaces via pupil modulation”, *J. Phys. Photonics*, **8**(3), 035011 (2026).
- [3] N. Chen[†], Y. Wu[†], **Chao Tan**, L. Cao, J. Wang, and E. Y. Lam, “Uncertainty-aware Fourier ptychography”, *Light: Sci. Appl.*, **14**(1), 236 (2025).
- [4] F. Xu, Z. Wu, **Chao Tan**, Y. Liao, Z. Wang, K. Chen, and A. Pan, “Fourier ptychographic microscopy 10 years on: a review”, *Cells*, **13**(4), 324 (2024).
- [5] **Chao Tan**, J. Wang, Y. Wu, J. Zhou, and N. Chen, “Fast scaled cylindrical holography based on scaled convolution”, *Displays*, **81**, 102619 (2024).
- [6] W. Zhang, J. Wang, **Chao Tan**, Y. Wu, Y. Zhang, and N. Chen, “Large field-of-view holographic Maxwellian display based on spherical crown diffraction”, *Opt. Express*, **31**(14), 22660–22670 (2023).

Conferences

- [1] **Chao Tan**, M. Jang, “Robust Fourier ptychographic tomography via feature domain”, *SPIE Advanced Biophotonics Congress (SPIE ABC)*, 2025.

Experience

Graduate Researcher: Computational Microscopy

Aug 2024 – Aug 2026

Mooo Research Group, KAIST

Daejeon, Korea

- Advanced research on 3D computational imaging by implementing and comparing algorithms for intensity diffraction tomography (IDT) and Fourier ptychographic tomography (FPT)
- Independently designed and built an off-axis holographic microscope

Research Intern: Computational Microscopy

Feb 2024 – Jun 2024

Information Display Institute, Sichuan University

Chengdu, China

- Contributed to the development of a novel differentiable reconstruction framework for FPM, which automated parameter tuning and improved reconstruction fidelity

Research Intern: Computational Microscopy

Jul 2023 – Jan 2024

Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences

Xi'an, China

- Authored the 3D imaging section of a comprehensive review on Fourier ptychography

Research Intern: Computer-generated Holograms

Jul 2022 – Jul 2023

Information Display Institute, Sichuan University

Chengdu, China

- Proposed a fast calculation algorithm for scaled cylindrical holography using scaled diffraction in Python

Skills

Research: Computational imaging, event-based vision, computational photography, computational microscopy, 3D imaging

Languages: English – Fluent (IELTS 7), Mandarin – Native speaker

Programming Languages: Python, MATLAB, LaTeX

Software & Tools: Zemax, SolidWorks, Blender

Hardware: Raspberry Pi